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Human Performance Capability in Psychomotor Tasks at Variable Difficulty Levels and Physiological Reactions under Noise and Heat Conditions

Sunil K. BHATTACHARYA*, Chandan K. PRADHAN, Somnath R. TRIPATHI and Shrikant KASHYAP

National Institute of Occupational Health, Meghani Nagar, Ahmedabad 380016, INDIA

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Abstract: The effects of noise and heat on the efficiency of human performance of psychomotor tasks at variable difficulty levels were investigated. Twenty four young, healthy volunteers with normal hearing, randomly divided into 4 equal sized groups, were tested in a climatic chamber under 4 combinations of two environmental conditions made up of noise and heat, each at two levels: 70 dB (A) (quiet), 100 dB (A) (noise): 28°C (comfortable), 35° C (heat). A statistically balanced procedure was followed. Three physiological variables were also recorded: Heart rate, blood pressure and body temperature. A 2×2 analysis of variance showed rise in the physiological variables by noise and heat respectively as well as by their interactions. Performances of the psychomotor tasks were affected differently by noise and heat depending on the difficulty levels of the tasks. The interaction of noise and heat entailed superior performance of the tasks of memory and search, two hand coordination and reaction time, at moderate difficulty levels, but no distinct interaction effect was observed on the performance of tweezer dexterity.

Key words: Heat—Noise—Physiological variables—Psychomotor tasks—Task difficulty levels

INTRODUCTION

Noise and heat, amongst many other factors, are concomitant stressors that produce complex work environment in shop floors of industrial processes. Yet, most of the investigations have been designed to understand human performance efficiency accompanied by noise or heat independently from each other. Hamilton et al[1] found that noise aided recall involving order information, while its detrimental effects have been reported by Broadbent and Gregory[2] on signal

* To whom correspondence should be addressed.

detection tasks and by Eschenbrenner[3] on sensorimotor tasks. Moreover, Ohwaki[4] reported no change in human performance by intermittent noise of 60 and 80 dB (A). In contrast, Poulton and Kerslake[5] showed an improvement in visual scanning of signal scores while the subjects were listening to auditory stimulus in hot environmental conditions. Our earlier investigation[6] on the effects of noise and heat on a few tasks showed decreased performance accuracy in cognitive tasks and improvement in performance of perceptual motor tasks by noise of 100 dB (A) in the work room at the air temperature of 35°C. The present investigation was, therefore, designed to determine the effects of noise interposed with heat on performance efficiency of psychomotor tasks at variable difficulty levels as well as on some physiological variables as indicators of activation level.

Materials and Methods

Subjects: A group of 24 male college student volunteers, having sound physical health with normal hearing, participated in this experiment. All of them wore cotton clothes providing insulation value of 0.6 clo unit[8]. They did not have any previous experience of exposures to noise and heat routinely. The anthropometric measurements of the subjects are given in Table 1.

Environmental measures: A few equipments such as audiometer (M/s. Arphi India Ltd, India), sound level meter, Type 2218 (B & K, Denmark), and a climatic chamber (M/s. Hotpack corporation, U. S. A.) were used. The details of the equipments are given in a previous report by Bhattacharya *et al*[6].

Physiological measures: Heart rate counter (Hauer, Switzerland), Deep body thermometer (DBT Ltd., England) and sphygmomanometer (Erkameter, Germany) were used for measurements of the physiological variables[6].

Heart rate counter: The heart rate of the subject was recorded by counting 10 heart beats with the help of a stethoscope (Reister, Germany) placed at the left fifth intercostal space of his chest, noting the time from the watch dial of the counter and then reading the appropriate scale in it[7].

Deep body thermometer: A deep body thermometer was used with its electrode taped on the chest of the subject to record body temperature (°C). The remote sensing equipment was kept outside the chamber.

Sphygmomanometer: The sphygmomanometer was kept at the level of the heart and the cuff was tied round the upper arm. The chest piece of the stethoscope was placed on the brachial artery. Pressure was raised to 200 mm Hg and then gradually released. The first clear occurrence of the tapping sound indicated systolic pressure while the just disappearance of the sound recorded diastolic pressure.

Psychomotor measures: A battery of psychomotor tasks containing tweezer dexterity, memory and search (letter cancellation), two hand coordination and

reaction time were used in this experiment. The details have been given by Bhattacharya *et al*[9, 10]. The scoring of the tasks were as follows:

Tweezer dexterity: The number of metallic pins inserted into the holes on wooden boards with the help of a tweezer in 3 min. The task was in two categories. In 'a' category the diameter of each pin was 3.5 mm and that of each hole 4.0 mm, while in 'b' category the diameter of each pin and hole were 1.5 mm and 1.6 mm respectively.

Memory and search task: The number of letters cancelled in an array containing 5 lines with 50 randomised letters in each line in 3 min. The three categories of the task were cancellation of: (a) one letter, (b) three letters and (c) five letters.

Two hand coordination: The time taken (sec) in tracing a geometrical pattern in two categories: (a) a straight line with deflections on its left and right ends, (b) curvatures in addition to that of 'a'.

Reaction time: The average time (in 1/100 sec unit) taken for 30 trials to respond to (a) one stimulus and (b) three stimuli as quickly after its onset as possible.

In all the tests, 'a' category represented the difficulty levels as low, 'b' category moderate and 'c' category high. The higher scores indicated superior performance in tweezer dexterity and memory and search task, while lower scores reflected better performance in two hand coordination and reaction time tests.

Design of the experiment: Based on the experience gained through surveys of noise and thermal environment in textile industries in India the levels of sound pressure which correspond to 70 dB (A) and 100 dB (A) and air temperatures which correspond to 28°C and 35°C respectively were administered.

In the climatic chamber wihte noise was generated by the audiometer, amplified through a set of speakers and measured with the sound level meter so that the sound pressure reached the desired level at the subject's ear.

The subjects were divided into four groups of 6 each, following a random number table[11]. First, under quiet condition (70 dB (A)), the subjects of group 1 worked at comfortable air temperature (28°C) and on the second occasion under not condition (35°C), while those of group 2 first worked under hot condition and on the second occasion under comfortable condition so as to eliminate the effects resulting from practice[11]. Likewise, under noise condition the subjects of group 3 first worked under comfortable air temperature and then on another occasion under hot condition while those of group 4 worked first under hot condition followed by comfortable condition on the second occasion. Thus each subject was tested under two different combinations of noise and heat, each lasting for an hour and half. The air velocity in the climatic chamber was 0.25 m/sec with the relative humidity being 50%. The thermal comfortable range of Indian subjects wearing cotton clothes of 0.6 clo unit was assessed to lie within a range of 23.3–28.3°C[12].

Procedure: The subjects were tested individually, at the same time of day (from 10.00 A. M.) for two days. They were briefed about the purpose of the experiment. Queries were clarified, if any. Cold or hot drinks were not allowed to the subjects during the experiment period. They were given the option to discontinue participation at any stage of the experiment, if they so wished. However, no subject loss occurred.

They were rested in a chair for 20 min for adaptation to each experimental condition following which the physiological variables were recorded.

The psychological tests placed on the table were administered according to standard instructions in the order in which this had been drawn up into a schedule as explained earlier and scored as per standard keys. The subjects were given rest for 5 min before administration of each psychological test and instructed to work as fast as possible without making error. A practice run was arranged for all the tests on the day before the test day. They were also informed fully of their performance scores.

Results

The test scores of the subjects, both psychological and physiological, were

Teble 1. Anthropometric measurements of the subjects (Values are mean±S. D.)

Group	Age (Yr)	Body height (cm)	Body weight (kg)	Body surface area (m^2)	Lean body mass (kg)
1	22. 66±3. 42	163. 25±4. 10	50. 28±5. 70	1. 52±0. 08	46. 07±5. 07
2	23. 83±4. 38	161. 83±3. 65	51. 05±4. 91	1. 53±0. 07	45. 98±4. 63
3	23. 17±4. 10	163. 91±3. 43	52. 75±5. 18	1. 53±0. 08	46. 08±4. 96
4	22. 67±4. 06	163. 17±4. 18	50. 0±5. 26	1. 52±0. 09	46. 57±4. 64

Table 2. Summary of analysis of variance of the physiological variables

Source of variation	df	Physiological variables							
		Heart Rate		Blood pressure (Systolic)		Blood pressure (Diastolic)		Body temperature	
		MS	F	MS	F	MS	F	MS	F
Between Subjects	23								
Noise effect	1	317. 30	4. 32*	28. 52	0. 28	140. 09	2. 14	0. 19	0. 48
Error	22	73. 45		99. 15		65. 35		0. 40	
Within Subjects	24								
Heat effect	1	320. 33	6. 26*	9. 19	0. 52	140. 09	2. 72	1. 78	7. 73**
Interaction effect of Noise with Heat	1	140. 09	2. 73	77. 07	4. 33*	70. 07	1. 36	1. 01	4. 41*
Error	22	51. 16		17. 80		51. 44		0. 23	

df: Degrees of freedom
MS: Mean square F: Fisher ratio
* $p<0.05$ ** $p<0.01$

Table 3. Summary of analysis of variance on tasks of tweezer dexterity and memory and search

Source of variation	df	Tasks[a]									
		Tweezer dexterity				Memory and Search					
		'a' Category		'b' Category		'a' Category		'b' Category		'c' Category	
		MS	F	MS	F	MS	F	MS	F	MS	F
Between Subjects	23										
Noise effects	1	70. 09	8. 09**	117. 19	17. 46**	2. 08	0. 19	9. 19	0. 56	30. 09	2. 30
Error	22	8. 66		6. 71		10. 54		16. 24		13. 08	
Within Subjects	24										
Heat effect	1	36. 75	3. 08	20. 02	2. 73	0. 33	0. 12	72. 52	11. 62**	24. 09	2. 51
Interaction effect of Noise with Heat	1	5. 33	0. 44	7. 52	1. 02	0. 75	0. 29	67. 69	10. 84**	37. 74	3. 93
Error	22	11. 90		7. 31		2. 58		6. 24		9. 59	

MS: Mean square; F: Fisher Ratio ** $p<0.01$ df: Degrees of freedom
[a] Difficulty levels of tasks in 'a' Category: Low 'b' Category: Moderate 'c' Category: High

Table 4. Summary of analysis of variance on tasks of two hand coordination and reaction time

Source of variance	df	Tasks[a]							
		Two hand coordination				Reaction time			
		'a' Category		'b' Category		'a' Category		'b' Category	
		MS	F	MS	F	MS	F	MS	F
Between Subjects	23								
Noise effect	1	5963. 02	7. 59*	2133. 33	0. 81	157. 88	13. 85**	273. 85	14. 12**
Error	22	784. 71		2627. 68		11. 37		19. 39	
Within Subjects	24								
Heat effect	1	3088. 02	2. 66	39736. 73	1. 58	166. 39	8. 63**	91. 44	5. 11*
Interaction of Noise with Heat	1	2366. 02	2. 03	108963. 63	4. 36*	26. 48	1. 37	77. 45	4. 33*
Error	22	1160. 24		24991. 66		19. 26		17. 88	

MS: Mean square; F: Fisher Ratio * $p<0.05$; ** $p<0.01$ df: Degrees of freedom
[a] Difficulty levels of tasks in 'a' Category: Low 'b' Category: Moderate 'c' Category: High

subjected to a two factor (2×2) analysis of variance with repeated measures design[11]. The two factors were: noise at two levels: 70 dB (A) (quiet) and 100 dB (A) (noise); and heat at two levels viz. 28°C (thermally comfortable) and 35°C (high heat). The results are given in Tables 2–4.

Table 2 suggests that there were distinct independent effects of noise and heat on heart rate but the interaction effect of these two was absent, as reflected by the F–values. On the other hand, the independent effects of noise or heat on blood pressure (systolic) was statistically insignificant, but the interaction effect of the stressors was appreciably high (as per F–value). However, there was no significant effects of noise and heat independently or jointly on blood pressure (diastolic). On body temperature, the F–value for heat was significant but not for noise. However, the combined effect of noise and heat was of considerable concern as shown by the significant F–value.

On the psychomotor tasks, Table 3 reveals that the F–values reached statistical significance for both 'a' and 'b' categories of tweezer dexterity test under noise condition. In memory and search task the rate of letter cancellation per min was computed. Noise did not have any significant effect on performance on this task but the heat had in its 'b' category. Also in 'b' category of the task the F–value for noise and heat interaction was found to be statistically significant.

Table 4 indicates significant F–value for noise on 'a' category of two-hand coordination, while that for the combined effect of noise and heat was insignificant. However, in 'b' category of the task noise influenced the performance considerably at high heat as reflected by the significant F–ratio. In reaction time the independent effects of noise and heat was remarkably noticeable. Also in 'b' category of the task the interaction effect of noise and heat was significantly high.

Discussion

It has been observed that the effects of environmental noise and heat on human performance in the form of either improvement, impairment or no change depend on the task, the particular combination of environmental conditions and the length of exposure[13]. The psychological and physiological variables studied in this investigation were based on the recommendation of ILO dealing with behavioural disturbances due to exposure to noise and heat[15].

The significant F–values for the independent effects of noise and heat on heart rate suggest that noise singly caused elevations to 83.50 beats/min from 78.01 beats/min and heat to 83.95 beats/min from 78.25 beats/min. On blood pressure (systolic), Fig. 1 illustrates appreciable interaction effect of noise and heat signifying that under quiet condition blood pressure increased with the rise in air temperature, which was depressed by noise in the presence of heat. Body temperature was affected by heat rising to 37.12°C at air temperature of 35°C from 36.48°C at comfortable condition. The combined effect of noise and heat

revealed (Fig. 1) that heat caused the rise of body temperature, which was further aggravated by noise.

On tweezer dexterity task noise caused deterioration in performance (mean 30.12) from that (mean 32.54) at quiet condition, in 'a' category while in 'b' category the mean performance decreased to 28.25 at noise condition from 33.97 at quiet condition. Thus, the degree of deterioration (15%) was more pronounced in 'b' category compared to that in 'a' category (7%). The most probable explanation is that the difficulty level in 'b' category of the task was higher in view of the precision coordination required between eye and hand compared to that in 'a' category of low difficulty level. Greater deterioration in this task at 'b' category of moderate difficulty level has resulted because noise increased heart rate which in turn increased blood supply to the tissues affecting physiological balance of the subjects[16].

The significant effect of heat in the 'b' category of memory and search task revealed degradation in performance (mean 12.75) at 35°C (high heat) from the

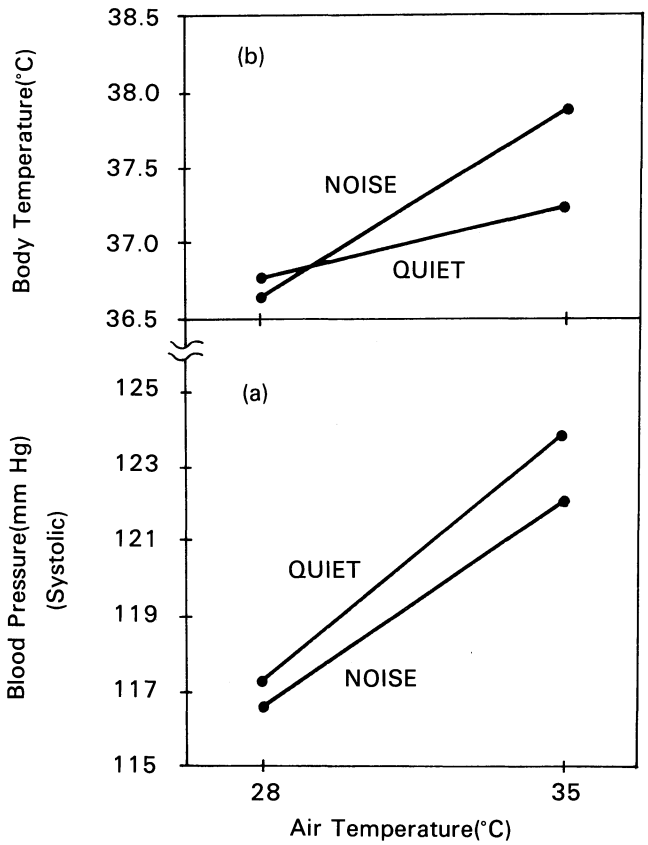


Fig. 1 Interaction effects of noise and heat on: (a) blood pressure (systolic) and (b) body temperature.

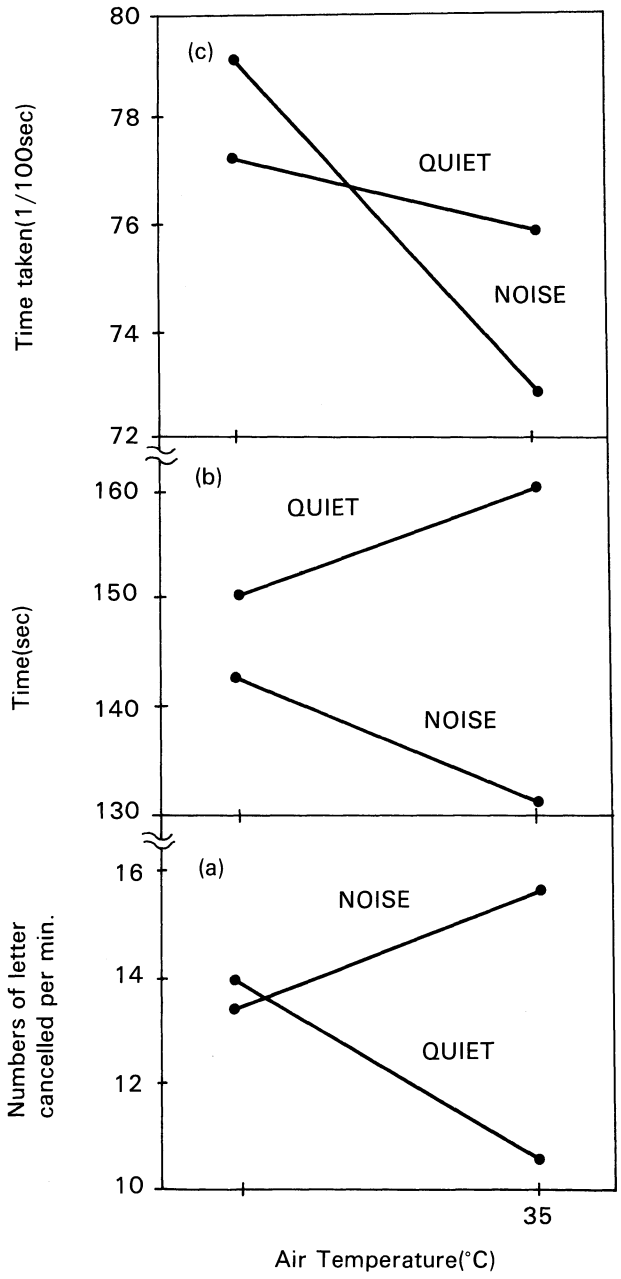


Fig. 2 Interaction effect of noise and heat on the tasks of: (a) memory and search, (b) two hand coordination and (c) reaction time, at moderate difficulty levels.

performance (mean 13.70) at comfortable condition (28°C). This has been because heat increased body temperature which is potent to disturb central nervous system functioning[17]. It also increased heart rate producing cerebral anoxia due to which the subjects failed to think in a decisive manner[18]. The interaction of noise and heat has been reflected in Fig. 2a showing performance depression with the rise in air temperature under quiet condition, while under noise condition the performance had beneficial changes at air temperature of 35° C. Such performance effects have been possible due to arousal effect of noise[19, 20] at high heat[13].

The independent effect of noise on 'a' category of two–hand coordination task was one of deterioration (mean 113.45) under noise condition compared to the performance (mean 91.16) under quiet condition. In this particular task noise acted as a dearousing stressor[3] lowering efficiency in coordinating the two hands of the subjects in tracing a less difficult geometrical pattern containing a straight line with deflections on its two ends. The interaction effect (Fig. 2b) reflected that performance deteriorated at high heat of 35°C under quiet condition, which improved under noise condition at raised air temperature of 35°C. As this category of the task was relatively difficult, noise in combination with heat possibly enhanced the arousal level of the subjects allowing higher order mental input and thereby resulting in favcurable performance.

Further in 'a' category reaction time was longer (mean 35.50) under noise condition compared to that (mean 31.89) in quiet condition, while in 'b' category, a reverse trend of shorter reaction time was observable, with the mean values being 74.98 and 76.54 respectively. Also, in 'a' category the heat prolonged the reaction time to the mean value of 35.57 from the mean value of 31.84 at comfortable condition, but in 'b' category reaction time was shorter to 74.42 at 35°C from 78.10 at 28°C. The reason for appearance of this phenomena was that noise or heat was able to produce arousal level in the subjects high enough to effect favourable performance changes on the more difficult task which is in agreement with the welford's[20] concept of arousal and task demand. Fig. 2c, illustrates that in 'b' category heat caused performance improvement under both quiet and noise conditions, but in the latter condition higher degree of improvement was noticeable. This shows that heat accompanied by noise raised the arousal level of the subjects, yielding superior performance effects.

It may thus be summarised that noise and heat caused elevations in heart rate and heat escalated the body temperature. The interaction effect of noise and heat also increased blood pressure (systolic) and body temperature. Adverse effects of noise was observable on tweezer dexterity task with higher degradation being at its moderate level, as well as on two hand coordination task at low difficulty level, while that of heat on memory and search task at moderate difficulty level. Noise and heat independently increased reaction time at low difficulty level but shortened at moderate difficulty level. The interaction of noise and heat pro-

duced beneficial changes in performance of the tasks of memory and search, two hand coordination, and reaction time at moderate difficulty levels.

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